Deep Learning and Its Application

Week 7: Practicals

1. Write a code in python, use a random way to generate a two dimensional matrix for grid processing. The matrix is composed of 10 rows and 10 columns, where the entries of the matrix is either 0 or 1.

2. Below is a formula that is used to compute the convolution between two same dimension matrices:

$$\begin{bmatrix} a & b \\ e & f \end{bmatrix} * \begin{bmatrix} w & x \\ y & z \end{bmatrix} = [aw + bx + ey + fz]$$

Given an input A = [[2, 3], [3, 4]], B = [[1, 2], [3, 3]], write a Python code to conduct their convolution calculation.

3. Below is a formula that is used to compute the convolution between two same dimension matrices:

$$\begin{bmatrix} a & b \\ e & f \end{bmatrix} * \begin{bmatrix} w & x \\ y & z \end{bmatrix} = [aw + bx + ey + fz]$$

Given an input A = [[2, 3], [3, 4]], B = [[1, 2], [3, 3]], write a Python code to conduct their convolution calculation using the dot product function of the array.

Tip: please refer to the Convolutional Operation given in Lecture 7

4. Define a one-dimensional input that has eight elements all with the value of 0, with a two element bump in the middle with the values 1 below

   [0, 0, 0, 1, 1, 0, 0, 0]

   Define a single kernel below

   [0, 1, 0]

   The following is a fragment of code for detecting the bump of the one-dimensional input above and complete the tasks below

```
from numpy import asarray
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv1D
# define input data
data = asarray([0, 0, 0, 1, 1, 0, 0, 0])
data = data.reshape(1, 8, 1)
# create model
model = Sequential()
model.add(Conv1D(1, 3, input_shape=(8, 1)))
# define a vertical line detector
```

```
weights = [asarray([[[0]],[[1]],[[0]]]), asarray([0.0])]
# store the weights in the model
model.set_weights(weights)
```

a) Use 'model' to get the stored weights and print them out
b) Input [0, 0, 0, 1, 1, 0, 0, 0] into the model and make a prediction
c) Print the predicted result
d) Modify the sizes of input and Kernel to see what will be predicted

Tip: please refer to the example of 1D Convolutional Layer given in Lecture 7

5. Read Chapter 4: Convolution Neural Networks at https://learning.oreilly.com/library/view/deep-learning-with/9781838823412/Text/Chapter_04.xhtml

   which requires to input your email along with password to login